

Car Fuel Efficiency Prediction Using Machine Learning

Submitted by:

B6737139 - Patrick Oliver Montano

B6741068 - Paing Nyein Oo

B6737115 - Sokvisal Leng

1. Background and Significance

Fuel efficiency is an important factor in vehicle performance, transportation cost, and environmental impact. A car with higher fuel efficiency consumes less fuel to travel the same distance, which reduces fuel expenses and lowers fuel consumption. Fuel efficiency is commonly measured using Miles Per Gallon (MPG), which represents how many miles a vehicle can travel using one gallon of fuel.

In real-world vehicle analysis, MPG is affected by several characteristics, including cylinders, engine displacement, horsepower, vehicle weight, acceleration, model year, and origin. These variables do not always affect MPG in a simple linear way. For example, a heavier car with a larger engine usually has lower MPG, but model year and vehicle origin may also influence the result because of differences in technology and manufacturing design.

Machine learning is useful for this problem because it can learn patterns from historical vehicle data and use those patterns to estimate MPG for new vehicles. This project uses the Auto MPG dataset and compares several regression models to predict car fuel efficiency.

2. Objectives

The main objective of this project is to develop a machine learning model that predicts car fuel efficiency based on vehicle characteristics.

The specific objectives are:

- To study the relationship between vehicle input features and MPG.
- To clean and prepare the Auto MPG dataset for machine learning.
- To train multiple regression models for MPG prediction.
- To compare model performance using MSE, RMSE, R^2 Score, and CV-RMSE.
- To identify the best-performing model for fuel efficiency prediction.
- To analyze the impact of each input feature on the predicted MPG.
- To test sample predictions using the trained models.
- To tune the SVR model hyperparameters and compare R^2 results.

3. Scope of the Project

This project focuses on supervised machine learning regression for car fuel efficiency prediction. The target variable is mpg, while the original input features are:

- cylinders
- displacement
- horsepower
- weight
- acceleration
- model year
- origin

The project includes dataset collection, data cleaning, preprocessing, training, evaluation, feature analysis, hyperparameter tuning, and sample prediction. It does not include real-time vehicle sensor data, modern fuel price prediction, mobile application deployment, or web deployment.

4. Methodology Based on the Steps of Machine Learning

Step 0: Determine Machine Learning Requirements

The first step was to define the machine learning problem clearly. The project is a regression problem because the output, MPG, is a continuous numerical value.

The machine learning requirements are:

Requirement	Description
Problem type	Regression
Target output	MPG
Input features	Cylinders, Displacement, Horsepower, Weight, Acceleration, Model Year, Origin
Dataset	Auto MPG dataset
Evaluation metrics	MSE, RMSE, R^2 Score, CV-RMSE
Main goal	Build a model that predicts MPG with low error and high R^2

Step 1: Gathering Data

The dataset used in this project is the Auto MPG dataset from the UCI Machine Learning Repository. It contains vehicle information and fuel efficiency values.

The original dataset columns are:

Column	Description
mpg	Fuel efficiency in miles per gallon

Column	Description
cylinders	Number of engine cylinders
displacement	Engine displacement
horsepower	Engine horsepower
weight	Vehicle weight
acceleration	Acceleration value
model_year	Vehicle model year
origin	Vehicle origin category
car_name	Vehicle name

The project code loads the dataset directly from the UCI source:

```
url = "https://archive.ics.uci.edu/ml/machine-learning-databases/auto-mpg/auto-mpg.data"
column_names = ['mpg', 'cylinders', 'displacement', 'horsepower',
                'weight', 'acceleration', 'model_year', 'origin', 'car_name']
```

```
data = pd.read_csv(url, names=column_names, na_values='?', sep=r'\s+', comment='\t')
```

After removing missing values, the dataset contained 392 usable records.

Step 2: Preparing the Data

The dataset was prepared before training the models. Missing horsepower values were represented by “?”, so they were converted into NaN and removed using `dropna()`. The `car_name` column was also removed because it is a text field and was not used as a numerical model input.

```
data = data.dropna().drop(['car_name'], axis=1)
```

```
X = data.drop(['mpg'], axis=1)
y = data['mpg']
```

The origin feature is categorical, so it was converted using one-hot encoding:

```
X = pd.get_dummies(X, columns=['origin'], drop_first=True)
```

After encoding, the model used these numerical input columns:

- cylinders
- displacement
- horsepower
- weight
- acceleration

- model_year
- origin_2
- origin_3

The original report still describes seven vehicle inputs because origin_2 and origin_3 are encoded versions of the single original origin feature.

Step 3: Choosing a Model

Several regression models were selected and compared:

Model	Reason for Use
Linear Regression	Simple baseline model
Ridge Regression	Linear model with L2 regularization
Lasso Regression	Linear model with L1 regularization
Elastic Net	Combines L1 and L2 regularization
Random Forest	Captures nonlinear patterns using decision trees
Gradient Boosting	Builds strong predictions from sequential weak learners
SVR with RBF Kernel	Captures nonlinear relationships using support vectors
Polynomial Regression Degree 2	Adds nonlinear polynomial feature interactions

The purpose of comparing different models was to identify which approach predicted MPG most accurately. The best test performance in the printed results was achieved by SVR with the RBF kernel.

Step 4: Training

The dataset was split into training and testing sets using an 80:20 split:

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

After cleaning, the dataset had 392 records:

Split	Number of Records	Percentage
Training set	313	80%
Testing set	79	20%

The 80:20 split was selected because it gives the model enough training data while still keeping a reasonable number of testing samples. Since the dataset is small, a 90:10 split gives a smaller test set and can make evaluation less reliable. A 70:30 split gives

more test data but reduces training data. Therefore, 80:20 is a balanced choice for this project.

The SVR split comparison was:

Train: Test Split	Test Samples	RMSE	R ²	CV-RMSE
90:10	40	1.859	0.9419	2.784
85:15	59	2.122	0.9231	2.772
80:20	79	2.225	0.9030	2.919
75:25	98	2.307	0.8945	2.901
70:30	118	2.358	0.8949	2.840

Although 90:10 produced a higher test R², it only used 40 test samples. The 80:20 split is more balanced and gives a more dependable evaluation for the report.

Feature Scaling

Feature scaling was applied using StandardScaler:

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

The purpose of scaling is to put all features on a similar scale. This is important because features such as weight and horsepower have much larger values than acceleration or origin. SVR, Linear Regression, Ridge, Lasso, and ElasticNet can be affected by feature scale, so scaling helps the model train more fairly and improves stability.

Model Training Code

The SVR model was trained using an RBF kernel:

```
from sklearn.svm import SVR
```

```
svr_model = SVR(kernel='rbf', C=100, gamma=0.1, epsilon=0.1)
svr_model.fit(X_train_scaled, y_train)
```

```
y_pred = svr_model.predict(X_test_scaled)
```

SVR Parameters Used in Training

Parameter	Value	Meaning
Kernel, K	RBF	Allows the model to learn nonlinear relationships

Parameter	Value	Meaning
C	100	Controls the penalty for prediction errors
gamma	0.1	Controls how far the influence of one training point reaches
epsilon	0.1	Defines the margin where small errors are ignored
bias term	23.4216	The fitted SVR intercept, stored in <code>svr_model.intercept_</code>

The RBF kernel function is:

$$K(x_i, x_{i'}) = e^{(-\gamma \|x_i - x_{i'}\|^2)}$$

Support Vectors

The trained SVR model used 297 support vectors out of 313 training samples, which is 94.89% of the training set.

A support vector is a training data point that directly affects the final SVR prediction boundary. In SVR regression, support vectors are usually the points outside the epsilon margin or on the margin. These points are important because the model uses them to form the prediction function. Points inside the epsilon margin have less influence because their prediction error is considered acceptable.

The support-vector count can be printed with:

```
print(len(svr_model.support_))
print(svr_model.support_vectors_.shape[0])
```

Step 5: Evaluation

The models were evaluated using MSE, RMSE, R² Score, and CV-RMSE.

Model	MSE	RMSE	R ² Score	CV-RMSE
Linear Regression	10.602	3.256	0.7923	3.445
Ridge Regression	10.656	3.264	0.7912	3.442
Lasso Regression	10.795	3.286	0.7885	3.429
ElasticNet	10.926	3.305	0.7859	3.471

Model	MSE	RMSE	R ² Score	CV-RMSE
Random Forest	5.691	2.386	0.8885	3.005
Gradient Boosting	6.702	2.589	0.8687	2.916
SVR with RBF Kernel	4.951	2.225	0.9030	2.919
Polynomial Regression Degree 2	7.657	2.767	0.8500	2.904

The SVR with RBF Kernel achieved the best test result, with:

Metric	Value
MSE	4.951
RMSE	2.225
R ² Score	0.9030
CV-RMSE	2.919

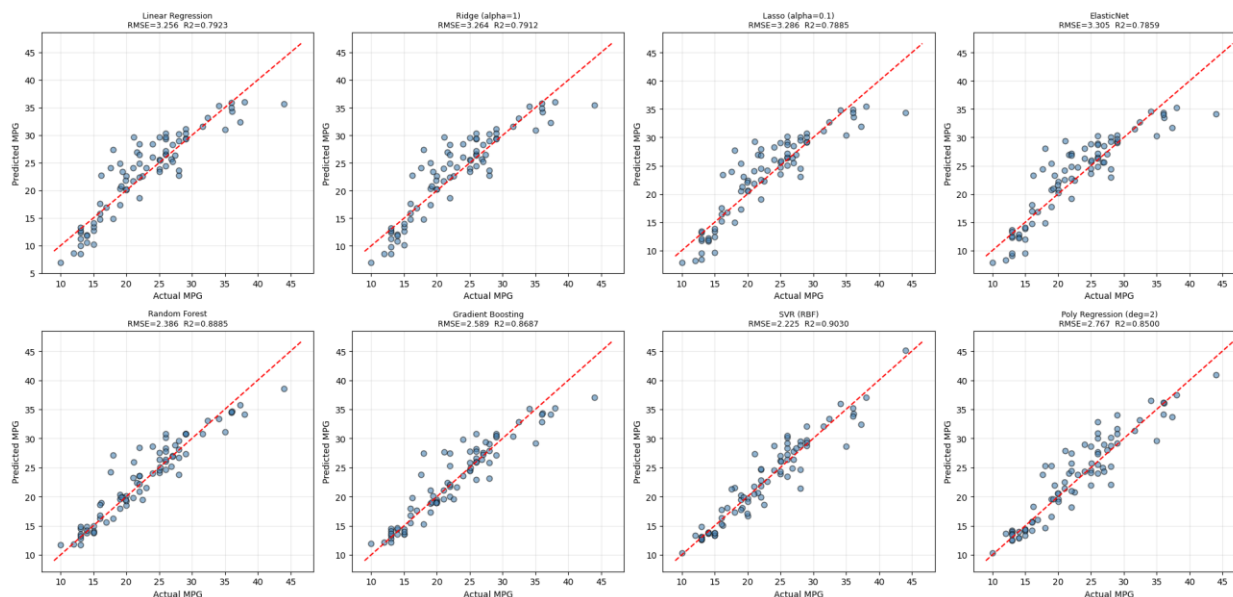
The R² Score of 0.9030 means the SVR model explained about 90.30% of the variation in MPG on the test set.

Actual vs Predicted Result

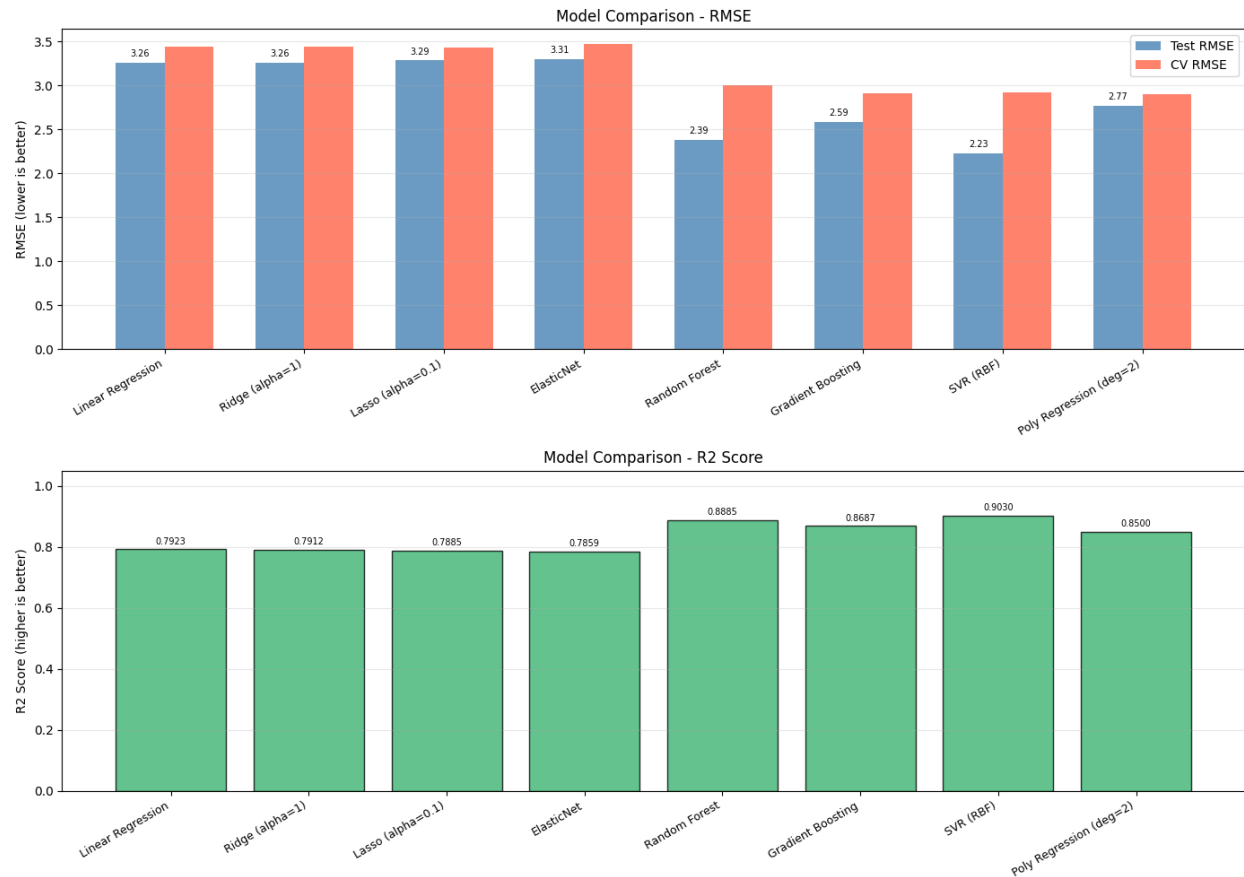
The actual vs predicted graph shows how close each model's predicted MPG values are to the real MPG values. Points closer to the red diagonal line represent better predictions.

Actual vs Predicted MPG - All Models

Actual vs Predicted MPG - All Models



The model comparison graph also shows that SVR had the lowest test RMSE and the highest test R².



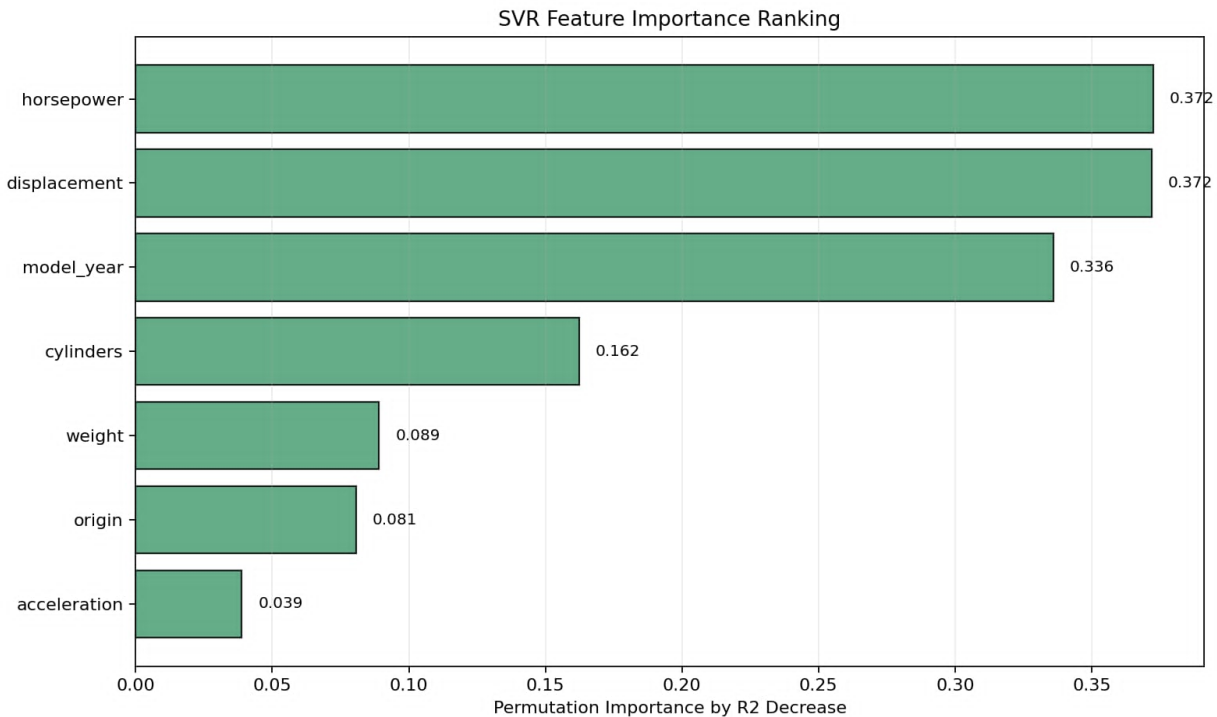
The observed relationships are:

Input Feature	Relationship with MPG
cylinders	Negative relationship. Cars with more cylinders usually have lower MPG.
displacement	Strong negative relationship. Larger engines usually reduce MPG.
horsepower	Strong negative relationship. Higher horsepower often means more fuel consumption.
weight	Strong negative relationship. Heavier cars usually have lower MPG.
acceleration	Weak positive relationship. Cars with higher acceleration values may show slightly higher MPG.
model year	Positive relationship. Newer model years tend to have better MPG.

Input Feature	Relationship with MPG
origin	Origin groups 2 and 3 show higher average MPG than origin group 1.

Feature Importance Ranking

Because SVR does not provide built-in tree-style feature importance, permutation importance was used. This measures how much the model's R^2 decreases when a feature is shuffled. A larger decrease means the feature is more important.



The SVR feature importance ranking was:

Rank	Feature	Importance
1	horsepower	0.3725
2	displacement	0.3719
3	model year	0.3359
4	cylinders	0.1623
5	weight	0.0891
6	origin	0.0806
7	acceleration	0.0388

The most influential features were horsepower, displacement, and model year. This is reasonable because engine power, engine size, and vehicle technology generation strongly affect fuel efficiency.

Step 6: Hyperparameter Tuning

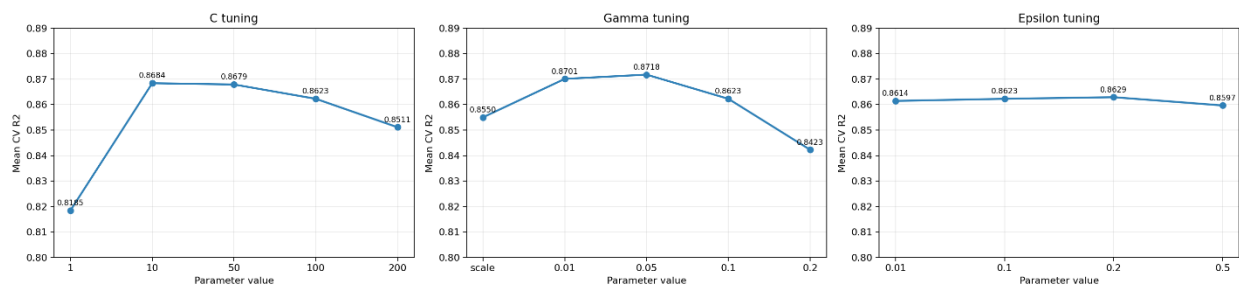
Hyperparameter tuning was performed for the SVR model using cross-validation R^2 . The main parameters tested were C, gamma, and epsilon.

Parameter	Purpose
C	Controls how strongly the model penalizes prediction errors
gamma	Controls the influence range of each training point in the RBF kernel
epsilon	Defines the no-penalty error margin around the prediction

The tuning graph is shown below.

SVR Hyperparameter Tuning Results

SVR Hyperparameter Tuning Results



Tuning C

The tuning test changed to C while keeping gamma=0.1 and epsilon=0.1.

C	Mean CV R^2
1	0.8185
10	0.8684
50	0.8679
100	0.8623
200	0.8511

The best single test for C was C=10, with a mean CV R^2 of 0.8684.

Tuning Gamma

The tuning test changed gamma while keeping C=100 and epsilon=0.1.

gamma	Mean CV R ²
scale	0.8550
0.01	0.8701
0.05	0.8718
0.1	0.8623
0.2	0.8423

The best single test for gamma was gamma=0.05, with a mean CV R² of 0.8718.

Turning Epsilon

The tuning test changed epsilon while keeping C=100 and gamma=0.1.

epsilon	Mean CV R ²
0.01	0.8614
0.1	0.8623
0.2	0.8629
0.5	0.8597

The best single test for epsilon was epsilon=0.2, with a mean CV R² of 0.8629.

Combined Grid Search Result

A combined grid search tested multiple values of C, gamma, and epsilon together. The best cross-validation result was:

C	gamma	epsilon	Mean CV R ²
50	0.05	0.5	0.8799

The top combined tuning results were:

Rank	C	gamma	epsilon	Mean CV R ²
1	50	0.05	0.5	0.8799
2	50	0.05	0.1	0.8772
3	50	0.05	0.2	0.8795
4	200	0.01	0.5	0.8754
5	100	0.01	0.5	0.8750

Based on cross-validation, the best tuned SVR parameters were:

SVR(kernel='rbf', C=50, gamma=0.05, epsilon=0.5)

This parameter set was selected by cross-validation because it had the highest mean CV R². However, the original SVR settings used in the printed project output, C=100, gamma=0.1, and epsilon=0.1, achieved the highest reported test R² of 0.9030. For a strict

machine learning workflow, the cross-validation result should be used to choose parameters, then the final model should be evaluated once on the test set.

The comparison between the original SVR and the tuned SVR was:

SVR Version	C	gamma	epsilon	Test RMSE	Test R ²	CV R ²	CV-RMSE
Original project SVR	100	0.1	0.1	2.225	0.9030	0.8623	2.919
Best CV-tuned SVR	50	0.05	0.5	2.323	0.8942	0.8769	2.779

The original SVR had the best test R² in the current project output, while the tuned SVR had the better cross-validation R² and CV-RMSE. Therefore, the final selected model for the report remains SVR with RBF Kernel, and the recommended tuned version for future retraining is C=50, gamma=0.05, and epsilon=0.5.

Step 7: Prediction

After training, the models were used to predict MPG for a sample vehicle.

The sample input was:

Feature	Value
cylinders	8
displacement	307
horsepower	130
weight	3504
acceleration	12
model year	70
origin	1

The sample prediction results were:

Model	Predicted MPG
Linear Regression	14.82
Ridge Regression	14.84
Lasso Regression	15.01
Elastic Net	14.88
Random Forest	16.27
Gradient Boosting	15.25
SVR with RBF Kernel	17.36
Polynomial Regression Degree 2	14.68

The final selected SVR model predicted:

17.36 MPG

5. Results Analysis

The results show that the simple linear models performed reasonably but were weaker than the nonlinear models. Linear Regression, Ridge, Lasso, and ElasticNet had RMSE values around 3.25 to 3.31 and R^2 values around 0.78 to 0.79. These models could identify general relationships, but they could not capture more complex nonlinear patterns in the data.

Tree-based models performed better. Random Forest achieved an RMSE of 2.386 and an R^2 Score of 0.8885, while Gradient Boosting achieved an RMSE of 2.589 and an R^2 Score of 0.8687. These models performed well because they can capture nonlinear interactions between features.

The best test result was achieved by SVR with RBF Kernel. It produced an RMSE of 2.225 and an R^2 Score of 0.9030. This means the model predicted MPG with the smallest test error and explained about 90.30% of the variation in MPG.

The relationship graphs show that cylinders, displacement, horsepower, and weight generally have negative relationships with MPG. This means larger, heavier, and more powerful vehicles usually consume more fuel. Model year has a positive relationship with MPG, which suggests that newer vehicles became more fuel efficient. Origin also has an effect, with origin groups 2 and 3 showing higher average MPG than origin group 1.

The feature importance ranking showed that horsepower, displacement, and model year were the strongest predictors for the SVR model. This supports the idea that engine power, engine size, and vehicle generation are major factors in fuel efficiency.

6. Conclusion

This project successfully developed and compared machine learning models for car fuel efficiency prediction. The Auto MPG dataset was cleaned, prepared, scaled, and used to train several regression models.

Among all tested models, SVR with RBF Kernel achieved the best test performance, with an RMSE of 2.225 and an R^2 Score of 0.9030. The model also predicted 17.36 MPG for the sample vehicle input.

The project showed that machine learning can effectively predict fuel efficiency using vehicle characteristics. The most important features were horsepower, displacement, and model year. In general, larger engines, higher horsepower, and heavier vehicles were associated with lower MPG, while newer model years were associated with better MPG.

Future improvements could include using a larger modern vehicle dataset, testing more advanced models, applying more detailed hyperparameter tuning, adding real-world vehicle features, and deploying the model as a simple web application.

References

UCI Machine Learning Repository. Auto MPG Dataset.
<https://archive.ics.uci.edu/ml/datasets/auto+mpg>